Integrating big data and AI for early lung cancer diagnosis in Jordan

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1. Introduction

Big data

Big data plays a vital role in each aspect of our modern lives. All Advanced technologies rely on integrating big data. Where big data provides information that helps reach a more informed and precise decision and predicts future trends and patterns which can be highly beneficial. The reason behind the effectiveness of big data is that a plethora of data is streaming because of the great volume of big data and the retrieval methods that are used to handle and analyze big data. Thus in most industries, we can notice that big data has a role that can not be overseen. Big data is more complicated than other data forms and has an enormous volume. Outpointing the 6'vs of big data will provide a comprehensive understanding of big data:

1. **Volume:**
   We can indicate from the name of big data that data can not be classified as big data unless it has a huge size, volume refers to the size of that data.
2. **Velocity:**
   Velocity refers to the speed of processing big data in order to get to a valuable output.
3. **Variety:**
   It reflects the nature of the data whether it is structured, unstructured, or semi-structured.
4. **Veracity:**
   It refers to the accuracy and reliability of the big data.
5. **Value:**
   Value characteristic outlines How beneficial big data is and what it provides the organization.
6. **Variability:**
   Which points to the big data usage and flow inconsistency, it doesn't directly describe big data but it highlights the importance of handling and managing it.

Big data in healthcare

Big data is the backbone of the healthcare industry, where the patients' records and information are stored as big data, where big data is analyzed and generated, to enhance the decision-making process, patient treatment, and many more daily procedures to attain a higher level of healthcare. The most common sources of big data in the medical field are EHR (electronic health records), EMRs (electronic medical records), and PHRs (personal health records). Big data offers a lot of opportunities in the healthcare sector, where big data analytics could be utilized for predicting the patient's diagnosis with the help of machine learning, resulting in early diagnosis and better treatment outcomes. Providing real-time notifications to the health care professional, helping in immediately taking action when changes in the patient's condition occur, and offering enhanced security for the patient's medical data which is crucial due to the sensitive nature of the data.

Utilizing Big data in the healthcare industry has many advantages such as attaining a higher level of patient treatment as it offers a better understanding of the patient's condition and experience due to the huge amount of patient data available. In addition, it offers better treatment plans by analyzing the previous treatment plans and which worked the best and which were not as good. As well as for its role in reducing the health care expenses on the patients and the hospitals, as big data can help in resource allocation, and early detection of possible healthcare issues early.

Big data and AI in lung cancer diagnosis

In recent years, cancer and especially lung cancer has become one of the main death causes worldwide. One of the main factors that help fight this disease and raise the effectiveness of the patient's treatment is early diagnosis and starting the treatment in the early stage of the disease, making it crucial to find early lung cancer detection technologies. Big data analytics and AI which revolve around making machines simulate human intelligence have played a vital role in resolving this worldwide health issue and saving millions of lives, as big data analytics has become a weapon in the battle against cancer during the past several years. Where these technologies have proved their effectiveness and efficiency in early diagnosis, detection, and treatment of lung cancer. Where AI is capable of generating and analyzing a significant amount of data in order to identify the symptom patterns that indicate the presence of lung cancer resulting in early diagnosis and more effective treatment plans.

Research objectives and questions

In this research, I aim to address and achieve these questions and objectives

Research questions:

- What would the early lung cancer diagnosis systems offer the Jordanian healthcare systems?
- What are the challenges that come along with integrating AI and big data analytics systems into the Jordanian healthcare sector?
- What are the ethical and legal concerns that should be considered when implementing these systems in Jordan?
- What is the suitable machine learning algorithm to be used in early lung cancer prediction?

Research objectives:

- To highlight the Jordanian community's ethical concerns and Jordanian laws that should be considered when implementing the system.
- To establish a basic guideline that could be further refined of how these systems would be implemented and utilized to help ease and make the diagnosis process more effective resulting in decreasing the number of deaths caused by lung cancer in Jordan.
- Evaluate the Potential Impact and Challenges associated with implementing an AI and big data-based early lung cancer diagnosis system in Jordan.

What makes my research different?

By setting these objectives and questions, I intend to investigate the benefits and challenges associated with utilizing AI and big data analytics for early lung cancer in Jordan. What makes my

research different is its focus on Jordanian hospitals and oncology departments and not focusing on suggesting AI algorithms but addressing the ethical and cultural concerns of the Jordanian community, technical challenges that might be faced in Jordanian hospitals, and Jordanian legal laws regarding implementing these systems as well. I will address these questions and objectives by Conducting a survey filled by specialists in oncology, data analytics specialists, and experts in the research field. In addition, testing machine learning algorithms to suggest a model that could be further refined for the early lung cancer diagnosis system will help address my questions and objectives as well. Aiming to help attain a higher level of cancer patient care and treatment in Jordan.

This Research content

The first section gives an introduction to the research topic in addition to the research objectives and questions in order to have a better understanding of the research aim. In the second section research papers related to my area of research will be reviewed to define the gaps that will be focused on. In the third section, the primary and secondary data used in this research will be defined. In the fourth section, the Onion model will be utilized, in addition to providing a flow chart describing how the research was conducted and completed. In the fifth section, the findings of the research will be summarized, in addition to highlighting how the findings align with the research objectives and questions. In the sixth section, the recommendations based on the research findings will be provided. The last section will be a reflection on multiple aspects of the research.

2. Related work and literature review:

In this literature review, I have used the Thematic structure where I started by reviewing research papers related to the difficulty of early diagnosing lung cancer, big data in healthcare, cancer diagnosis challenges and opportunities, and narrowing down to research papers related specifically to the use of AI and big data in early lung cancer diagnosis. This step will help ensure that the research objectives and questions are set correctly as reviewing past research papers will give me an overview of the gaps and what are the aspects that I should focus on in order to provide valuable and insightful research.

To have a better understanding of lung cancer diagnosis, what are the obstacles faced when early diagnosing lung cancer, and the importance of the early diagnosis I reviewed “The complexity and difficulty of diagnosing lung cancer: findings from a national primary-care study in Wales” (2015). This article discusses how diagnosing lung cancer can be complicated and challenging in primary care settings in Wales, the goal of this study is to have a better understanding of the challenges and difficulties faced when identifying the symptoms of lung cancer and its impact on early diagnosis. The study used a mixed method to conduct the research, where it used quantitative data from the national health records in addition to interviews with doctors who provided qualitative data. The results of the study show that lung cancer is difficult to diagnose in the early stages due to the unclear and non-specific nature of the early symptoms as they can be mixed up as common illnesses symptoms. Resulting in late diagnoses and treatment. Thus the author highlighted the need for enhancing diagnostic tools.
The article could have limitations as it has focused only on Wales and there could be bias from the doctors’ interviews.

“The complexity and difficulty of diagnosing lung cancer: findings from a national primary-care study in Wales” paper highlighted the need for improving lung cancer diagnosis tools. Thus I reviewed "Big Data in Healthcare - Opportunities and Challenges" (2018) article to explore the potential and challenges of integrating big data into healthcare to establish an overview of the benefits and drawbacks of using big data in enhancing early lung cancer diagnosis. where This research paper discusses the benefits and drawbacks of utilizing big data in healthcare. Where it highlights how big data could improve healthcare services and it outlines the challenges associated with utilizing big data in healthcare. The approach that the author followed to conduct this research is to review previous research papers related to big data and healthcare. The results show that the major opportunities of utilizing big data in healthcare are enhancing care quality, detecting fraud, lowering expenses and improving resource allocation, providing better treatment plans, and improving patient monitoring. On the other hand, utilizing big data in healthcare could raise privacy and security concerns which is a significant challenge, and there is a fear that it could lead to replacing the medical staff. There are weaknesses to this paper as the research scope is limited. In addition, the paper focuses more on the advantages and benefits of big data in healthcare while the challenges are not discussed enough.

To have a better understanding and overview of technologies that might improve early lung cancer diagnosis, I reviewed “The Clinical applications of artificial intelligence and machine learning in cancer diagnosis review” (2021) paper which gave a more specific overview of the challenges and opportunities of integrating Machine learning, AI and big data in cancer diagnosis. This paper comprehensively discusses the role of machine learning and AI in revolutionizing cancer treatment and early diagnosis by enhancing decision-making. The author didn’t mention the research strategy. The results closely discussed the applications of AI and its potential in analyzing big medical data such as image processing (radiology) where AI can detect any abnormal growth of cells and unjustified changes in the biological body which will result in early cancer diagnosis, as well as demonstrating its impact on precision oncology where utilizing AI for big datasets analysis will ease and raise the efficiency of providing personalized treatment plans, as the unique protein and genes of each tumor will be identified by the help of AI. The results addressed multiple concerns and challenges that come along with integrating AI into cancer diagnosis and oncology including data security where integrating machine learning for data analysis will require access to the patient's data, algorithm transparency as deep learning algorithms are frequently defined as “black box” thus the decision-making process is usually un-understandable for humans, AI algorithms can be bias which will cause fairness challenges, the need of appropriate frameworks where there should be a balance maintained between innovation in developing the AI algorithms and the patients’ safety, integration into the healthcare systems and ethical concerns related to the patient’s consent. The drawback of this paper is that it is not comprehensive and generalized where it doesn’t focus on a specific cancer type.

As the previous papers highlighted that there are multiple aspects that should be considered when implementing cancer diagnosis systems, it has given me an overview of what are the concerns that I should focus on addressing when conducting my research and what are the opportunities that will be provided when implementing such systems. To get more specific insights I started reviewing research papers that suggest early lung cancer diagnosis approaches in addition to underlying the challenges associated with these systems to highlight the gaps and what are the aspects that I should focus on. I have followed the Chronological structure where I started reviewing research papers from the last 4 years until the current year, aiming to stay up to date

with the latest technologies and get valuable insights into what are the best algorithms and approaches to follow when conducting my research.

"Particle Swarm Optimization Assisted Support Vector Machine Based Diagnostic System for Lung Cancer Prediction at the Early Stage", published on 02 November 2020. This article highlights the importance of early lung cancer diagnosis thus it suggests an approach to enhance early lung cancer diagnosis. it proposes the method of implementing a hybrid machine learning algorithm that combines the Particle Swarm Optimization (PSO) and Support Vector Machine (SVM) algorithms. Where the role of PSO is to filter the best parameters for the SVM classifier that is used for binary classification to diagnose the patient. In addition, the dataset used to train and test the model is lung cancer patients related, ensuring that the system is tested realistically. The results of this study show that there is a noticeable improvement In the accuracy score where it has reached 99% and the MSE score is lower compared to other models. The author didn't mention the research strategy, but the results were reached by evaluating the suggested mode. There were challenges mentioned in the article regarding the complexity of processing and handling the data. The author concludes the article by mentioning the need for further refining the model in order to enhance the model's performance. the method suggested in this article is considered effective and efficient, but there wasn't consideration of the fact that the PSO and SVM method highly relies on the data quality, as well as that this approach requires significantly robust processing power which could cause a significant challenge when implementing this approach in real life, indicating that this study lacks Real-World testing, which is crucial to ensure the effectiveness and efficiency of the method suggested.

Moving on to another advanced technique, the following study investigates another approach by using a deep learning algorithm.

"Artificial Intelligence–Based Prediction of Lung Cancer Risk Using Nonimaging Electronic Medical Records: Deep Learning Approach" published on 03 August 2021. This study focuses on predicting lung cancer using a convolutional neural network model referred to as Xception architecture depending on the patient's records including their information such as age and gender diseases they have been diagnosed with, what surgeries they had before, and prescriptions they had before. The researcher didn't mention a specific research methodology, but the results were reached by evaluating the suggested mode. The results of the suggested model show high accuracy where it was 0.90 for the overall population. The author mentioned some challenges associated with implementing these systems such as that the electronic medical records of the patients have a lot of features and details making it complex to analyze and generate. In addition, combining multiple types of data such as the diseases that the patient was diagnosed with and the medication he has been taking could cause challenges and decrease the accuracy if it wasn't carefully processed. The study didn't discuss the legal and ethical concerns that come along with integrating this system. In addition, the model was trained on Taiwan patient's medical records making it uncomprehensive where the risk factors of lung cancer could be different from one population to another.

While the first two research papers focus on utilizing machine learning and deep learning, the next paper suggests an approach using AI and liquid biopsy.

“A Classifier for Improving Early Lung Cancer Diagnosis Incorporating Artificial Intelligence and Liquid Biopsy” published on 03 March 2022. This study focuses on utilizing AI to predict the risk of lung cancer through the patients’ information such as smoking history, age, CT scan, and blood test. Where the method used was a three-dimensional U-net-based convolutional neural network on the patients’ blood tests and CT scans to classify their lung cancer risk. The researcher didn’t mention the research methodology, but the results were reached by evaluating the suggested model. The results showed that the suggested method provides high accuracy compared to other methods, where the model’s accuracy was 0.88. the author mentioned the challenges that might be associated with implementing this system such as integrating the system into the hospitals’ workflows and the need for extremely powerful computational resources. The study is conducted on the Chinese patients’ records which makes it uncomprehensive where the risk factors of lung cancer could be different from one population to another. In addition, the study mentions the challenges that might be faced when integrating the suggested AI approach but it doesn’t suggest strategies to address these challenges.

Moving on to the next paper that focuses on utilizing machine learning.

"Lung Cancer Prediction Using Machine Learning Technique Over Big Data," published on May 2022. The article highlights how crucial early lung cancer detection is, as the potential of early lung cancer diagnosis is low with a percentage of (14%), which has a significant impact on the effectiveness of the treatment. The article emphasizes the role of big data and machine learning in attaining a higher level of lung cancer patient treatment, as it will result in early cancer diagnosis. it discusses the importance of big data analysis in gathering valuable insights, in addition, it mentions that big data needs advanced algorithms where it can’t be processed using basic frameworks. In this study, Decision Tree & ID3, K-Nearest Neighbor (KNN), Random Forest, and Naive Bayes ML models are trained on numeric data only to predict lung cancer. The author didn’t mention the research methodology specifically, but the results were reached by evaluating and testing multiple models and analyzing the results. The results show that the Decision Tree & ID3 had the best performance with a high accuracy percentage (96.62%). Furthermore, the article discusses the challenges that could be faced when implementing the system such as the complexity of big data which was also mentioned in “Particle Swarm Optimization Assisted Support Vector Machine Based Diagnostic System for Lung Cancer Prediction at the Early Stage” research paper, where handling big data needs advanced software because of its huge volume and variety where in the field of healthcare data is gathered from multiple sources which causes a challenge. In addition, it mentions a concern regarding prediction accuracy where reaching high accuracy could be hard as well as choosing the appropriate models to be integrated. The study has multiple limitations that should be highlighted, where the training data only has numerical values making it inefficient to use in real hospitals. In addition, the study doesn’t address the integration and patients’ data privacy challenges associated with implementing this machine learning-based system into the existing real-life hospital frameworks.

Moving on to the last paper that suggests a deep learning and multi-omics data approach.

“Enhancing Lung Cancer Classification and Prediction with Deep Learning and Multi-Omics Data”, published on 26 April 2024. This paper suggests a deep learning approach for lung cancer prediction and classification, it utilizes the Convolutional Neural Network algorithm to

predict lung cancer from the lung cancer patients' data, especially lung adenocarcinoma which is a common lung cancer type. It focuses on multi-omics data which refers to a comprehensive examination of multiple "omics " which are the layers in a biological sample. The researcher didn't mention the research methodology, but the results were reached by evaluating the suggested model. The results of the study show that the CNN model has shown high and effective performance where its accuracy score was 0.97. The author mentioned some challenges that come along with Utilizing CNN for predicting lung cancer from multi-omics data lung cancer such as the need for carefully choosing the samples when combining multiple types of data such as mRNA and DNA to make sure of data reliability. Despite the effectiveness of the suggested neural network model, the author didn't discuss the challenges enough such as the ethical concerns, integration into the current hospitals' workflows and the complexity of handling big data.
The research papers related to utilizing AI and big data for early lung cancer diagnosis collectively highlight the role of machine learning and big data in enhancing early lung cancer diagnosis, emphasizing both the promising results and the challenges to be addressed when implementation.

Conclusion :

The first research paper discussed the complications and difficulties faced when diagnosing lung cancer in addition to its importance, where it highlighted the need for improving lung cancer diagnosis tools. This made me explore how to advance lung cancer diagnosis tools using big data and AI, thus to have a better understanding of how big data could benefit the healthcare sector and what concerns I should pay attention to I have reviewed a research paper discussing big data in healthcare. To even gather more valuable and detailed insights I reviewed a research paper that discussed AI and big data's impact on cancer diagnoses and what are the concerns that should be put into consideration. To gather more specific insights and explore the recent technologies suggested for early lung cancer prediction systems I have reviewed research papers specifically discussing my area of research. The five research papers suggested AI approaches and algorithms to early predict lung cancer. Despite the promising results of these methods and approaches implementing these systems requires a lot of careful preparation to overcome the challenges.

By reviewing the papers that discussed the challenges associated with implementing cancer diagnosing systems and big data impact on healthcare, and reviewing the research papers specifically discussing AI and big data approaches for implementing early lung cancer diagnosis systems. We can notice a crucial gap where in the research papers related to lung cancer diagnosis systems and approaches there wasn't enough consideration to the challenges associated with these systems that were discussed in the first three papers. In addition, most research papers limit their study to a particular population, and there are no research papers focusing on the Jordanian population. Indicating that There is a crucial need for further research in this field in order to fill the gaps that are causing obstacles to utilizing these AI algorithms.

In order to fill these gaps I aim to focus on the Jordanian population and hospitals in my study, by identifying the ethical, legal, cultural, and technical barriers associated with implementing these systems in Jordan along with suggesting an AI algorithm to early diagnose lung cancer.

3. Data Collection and Description

What is Primary data?

Primary data is the directly collected data by the researcher for the first time. Thus, the researcher has to supervise and start collecting the data from scratch, where the researcher has to manage and control the data collection process from the beginning till being satisfied with the data collected. There are many ways to gather primary data such as questionnaires, observation, and interviews. In my research, I have gathered data from questionnaires that were sent to specialists in my research area, where I have supervised the data collection from the beginning to the end, started by creating a survey that includes the questions that will help me answer and achieve my research questions and objectives, sending it to the specialists, analyzing the results and extracting valuable insights from the collected data.

What is secondary data?

Secondary data is the already collected data by other researchers, where the new researcher could use this data to extract new valuable insights that help him complete his research without consuming time and effort in the data collection process. My choice of secondary data in my research was a dataset already collected and well structured, the data set contained information about patients that could possibly have lung cancer such as their smoking history, symptoms, and daily habits.

Difference between primary and secondary data:

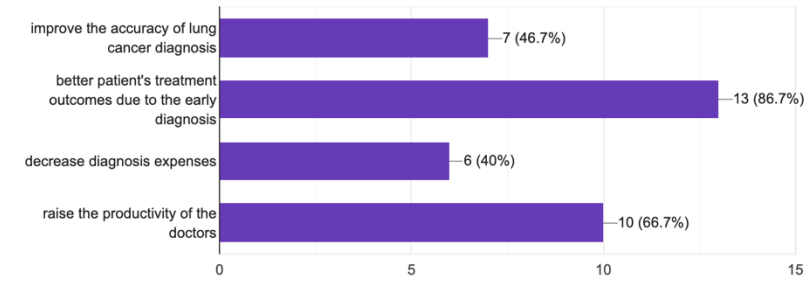
Data collection:
The primary data was collected for the first time by me while the secondary data is already collected by previous researchers and is ready to be used.
Time difference:
The primary data I collected is real-time data, while the secondary data is past data.
Data collection process:
Collecting primary data required time and effort as I was responsible for managing the data-gathering process while using secondary data didn't require effort and as it is ready to use.
Data form:
Primary data is in its raw form where I had to process it to extract valuable insights, while secondary data is already processed and ready to use.

3.1 Primary data:

My choice of primary data gathering was conducting a survey, that will be sent to specialists in the data science field, oncology doctors, Proctologist doctors, and healthcare data analysts, where the number of responses was 15. The survey included qualitative and quantitative questions that would help address and achieve my research questions and objectives comprehensively. In this section, I'll discuss the survey questions and the data gathered through this survey:
Survey link: https://forms.gle/pcteye2zRCKLZRxy8

From your perspective, how would integrating AI and big data analytics for early lung cancer diagnoses benefit the Jordanian healthcare industry?

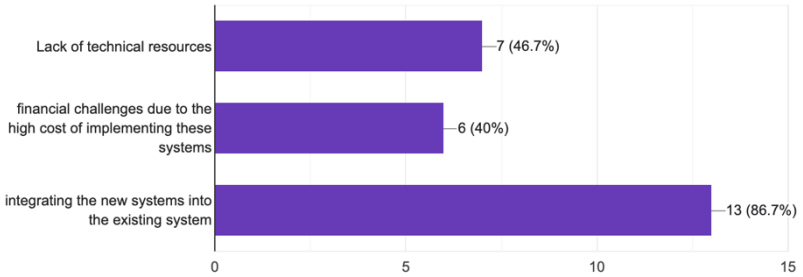
15 responses



Most of the responses indicate that the early Lung cancer prediction system will benefit the Jordanian healthcare industry by improving patient treatment outcomes due to early diagnosis and would raise the productivity of the doctors.

What are the challenges that could be faced when implementing these systems in Jordanian hospitals?

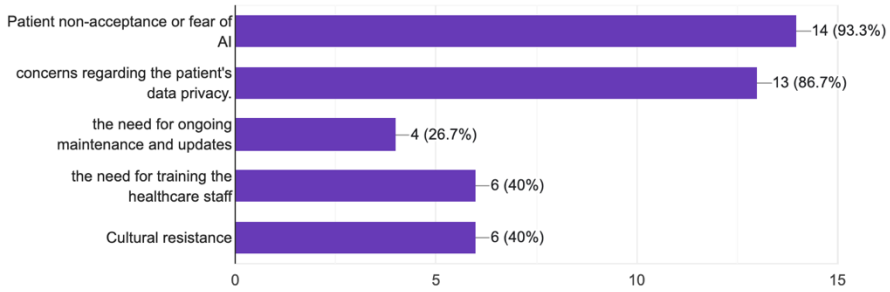
15 responses



Most responses indicate that the major challenge that would be faced when implementing the system would be integrating the new system into the existing system.

From your point of view what are the obstacles that could be faced after implementing this system related to adapting to the new system?

15 responses



Most responses indicate that the most likely obstacles to be faced after implementing the system is Patient non-acceptance or fear of AI and concerns regarding the patient's data privacy.

From your perspective, what are the ethical concerns that should be taken into consideration when integrating this system into Jordanian hospitals?

15 responses

- patient consent
- the patient's nonacceptance of being diagnosed by AI
- the need for taking the patient's consent
- Patient concent- for the doctor not to be biased with his diagnosis because of the results on the AI technology diagnosis, but to integrate both his medical opinion and the AI technology.
- patient consent and trust with the AI algorithm
- patient data privacy
  Bias in AI algorithms
  Consent for AI-based diagnosis
  Transparency in AI decision-making
- Patients should be informed that diagnosis is done with the help of AI.

From your perspective, what are the ethical concerns that should be taken into consideration when integrating this system into Jordanian hospitals?

15 responses

- Patients should be informed that diagnosis is done with the help of AI.
- the patient not approving for being diagnosed using machines and not human
- Consent of the patient
- being transparent with the patients about how their data will be used and how they will be diagnosed
- the transparency of how the decision was made
- the doctor's bias toward the AI prediction without making sure of the diagnosis
- culture acceptance of being diagnosed by AI
- patient consent and being transparent with the patient about how the decision was made
- Patient right to informed consent, the use of AI as a diagnostic tool instead of being used to assist with diagnosis.

Specialists were asked about what are the ethical concerns that should be taken into consideration when integrating this system into Jordanian hospitals, most of their responses were related to obtaining the patient's consent and informing them that their diagnosis was made by AI, in addition, some responses highlighted the concern of the doctor being bias to the AI diagnosis and not utilizing his expertise.

What are the Jordanian legal laws that should be taken into consideration before implementing these systems?

15 responses

- MInistry of health patient protection laws
- Doctor and Patient confidentiality
- electronic data privacy regulations
- Protecting the patient privacy
- Minister of Health approval and alignment with its regulations
- Data protection laws Medical liability regulations AI and machine learning governance Patient consent laws
- Data privacy laws.
- ministry of health approval of the system

What are the Jordanian legal laws that should be taken into consideration before implementi these systems?

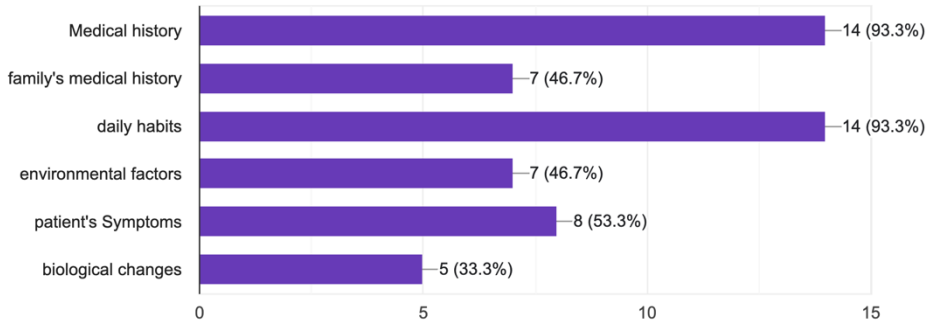
15 responses

- Data privacy laws.
- ministry of health approval of the system
- Insuraning patient privacy
- minister of health patients protection laws
- doctor patient confidentiality
- not my area of expertise
- minister of health and Jordan Food and Drug Administration laws
- - -
- Not my field of expertise

Specialists were asked about what are the Jordanian legal laws that should be taken into consideration before implementing the system, and most of their responses indicated that the Ministry of Health's legal laws, data privacy, patient protection, and patient-doctor confidentiality should be the first priority.

What are the criteria that should be followed in order to diagnose the patients?

15 responses



Most responses indicate that the criteria that the patients should be diagnosed depending on are their medical history and daily habits.

what are the technologies currently used for diagnosing lung cancer?

15 responses

- Imaging techniques
- Medications taken by the patient, blood tests, and imaging technologies
- CT-scans and blood test
- Radiology - Ct scan- MRI- tumor marker- biopsy- PET position emission tomography- lung function tedt
- imaging technologies and lung function test
- CT scans PET scans Chest X-rays Biopsies Blood tests
- -
- PET, tumor marker, blood test, and biopsy

what are the technologies currently used for diagnosing lung cancer?

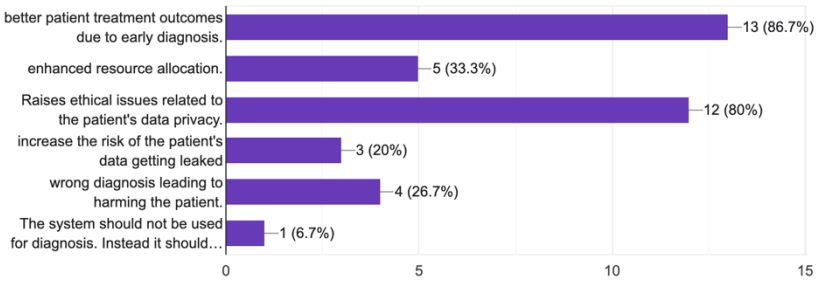
15 responses

- PET, tumor marker, blood test, and biopsy
- Tumor markers , CT , MRI , Biopsy , PET
- reviewing the electronic medical records to know the patient's medical history
- EMR
- biopsy , ct scan , MRI, lung function test and EMR to review the diseases the patient had and the medications he was on
- CT scan, X-ray and biopsy
- biopsy and patients medical history
- Imaging technology used in diagnosis of all types of cancer as well as blood and biopsy testing and imaging technologies used for all cancer types

Specialists were asked about the technologies currently used for diagnosing lung cancer, many technologies were mentioned as shown but the most mentioned are imaging techniques, blood tests, biopsies, EMR, and the patient's medical history including medications they took and diseases they were diagnosed with.

From your point of view, how would utilizing AI and big data analytics (the new system) in early lung cancer diagnosis impact the patients?

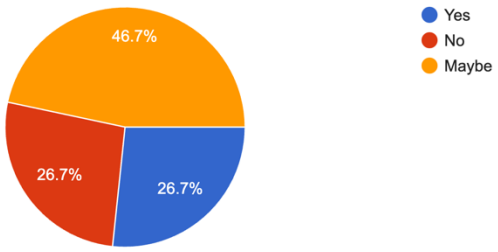
15 responses



Most responses agree that utilizing AI and big data for early lung cancer diagnosis will impact patients by enhancing their treatment outcome due to the early diagnosis and it would raise ethical issues related to the patient's data privacy.

Do you think that changing the currently used framework will cause an obstacle?

15 responses



if yes please justify your answer

4 responses

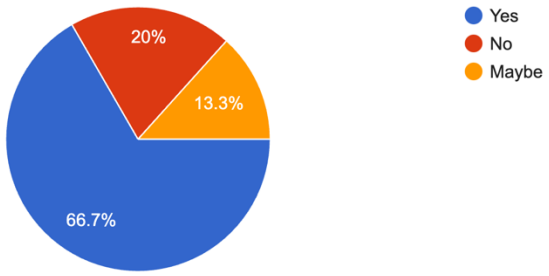
- It would take a long time transfering data and might lead to obstructing the work of doctors temporarily .
- the need for a robust mechanism for transferring the patient data to the new system
- losing the patient's data
- The framework took years of work and a lot of resources to develop and upending the whole system will be extremely costly

When the specialists were asked if they thought that changing the currently used framework would cause an obstacle 46.7% of them answered maybe, 26.7% answered no, and 26.7% answered yes. the specialists that answered yes justified their answer where they thought that it would cause an obstacle because it

could lead to losing the patient's data, it could have a high cost, and transferring the data would take a long time and obstruct the doctors.

Do you have concerns regarding the patient's privacy and data security?

15 responses



If yes from your point of view how can we address them?

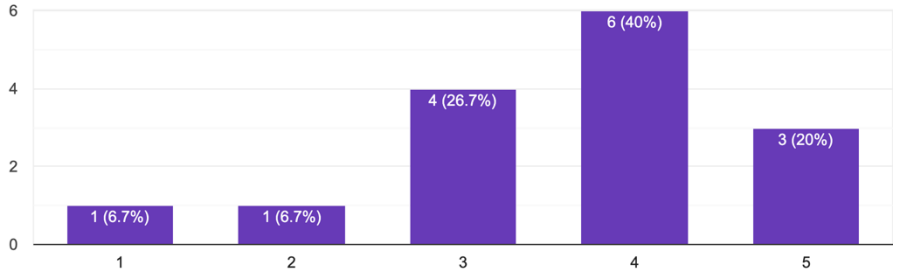
10 responses

- cybersecurity regulations and encryption
- patient data breach will cause legal issues
- Setting a very strict regulations on the staff to insure protecting it- setting punishments for leaking any data such as salary deduction .
- encrypting the patients data and only giving data access privilege to the individuals that need to
- Selling patients data to third parties for commercial reasons including insurance companies.
- strict regulations and Legal liability to ensure the patient's data privacyt
- by ensuring the presence of a powerful security system
- implementing a strict protection mechanism
- the first priority that should be considered before implementing the system is establishing a powerful and strict data protection regulations
- To ask for patient consent and make sure the consent is informed. The use of security measures to encrypt and protect patient data

The majority of the specialists have concerns regarding the patient's privacy and data security. When they were asked about how we can address them most of them agreed that a powerful patient data security system should be implemented. In addition, they suggested encryption, legal liability, and strict regulations to be set for the staff.

Do you have concerns regarding the accuracy of the system prediction?

15 responses



The responses indicate that there are high concerns regarding the system's accuracy.

please mention the concerns that you have regarding the accuracy of the system

15 responses

False negative will cause the patients to relax while the disease is progressing.

The system should not be used to diagnose patients but to asses risk of developing lung cancer. Diagnoses without the use of AI with years of human experience and knowledge are still inaccurate in some cases and AI is too underdeveloped to be used in this manner

if the patient was misdiagnosed with lung cancer and was harmed due to the unnecessary chemotherapy

if the patient wasn't diagnosed correctly the doctor and the hospital will be legally harmed

diagnosis must be reviewed by the doctors because any mistake will affect the doctors reputation significantly

false negative and false positive

please mention the concerns that you have regarding the accuracy of the system

15 responses

potential for false positives leading to unnecessary stress and procedures
Risk of false negatives resulting in missed diagnoses
Dependence on the quality of data input, which may be inconsistent
Algorithm bias affecting certain demographics disproportionately
Lack of transparency in AI decision-making processes
Challenges in validating and verifying AI predictions against current medical standards

the system should be well trained before integrating it into the hospital

false negative leading to the patient not getting treated when he needs to

I trust that the AI technology wont be implemented unless after a many and strict trials and studies and also a high accuracy percentage.

AI could have a percentage of inaccurate analysis and results

the need for continuous model training before using it in the hospitals to ensure the patients safety

the dr depending on the AI system, that could be not accurate in some cases resulting in harming and miss diagnosing the patient

false positive and false negative

AI hasn't reached the point where it could be completely reliable and it could miss diagnose patients which is not acceptable

The concerns related to the system's accuracy that the specialists had are false positive and false negative where if the patient was miss-diagnosed with lung cancer he could be harmed due to the unnecessary chemotherapy, on the other hand, if he was miss-diagnosed as lung cancer free his cancer could develop to advanced stages while he needed to be treated, AI still has not reached the point where it can be fully accurate and completely relied on and the concern of miss diagnosis by AI could cause legal liability and harm the reputation of the doctors and the hospital

Primary data gathering and challenges faced during the process:

The first step was deciding my approach for collecting the primary data I have chosen to conduct a mixed-method survey. The second step was writing down my survey questions, in this step I focused on making my questions straight to the point and comprehensive where they should address my research questions and objectives. The third step was to start sending the survey to the specialists, in this step, I paid close attention to how are the specialists to whom I should send the survey, as they have to be experts in my area of research in order to provide me with valuable insights.

While collecting the primary data I have faced challenges, as I struggled to find experts in my area of research and it took me time to reach out to them. In addition, I wasn't able to collect a lot of responses.

Justification for choosing the survey as my primary data-collecting approach:

I have chosen to collect my research primary data by conducting a survey as it enables me to find answers to the questions related to my research specifically, gives me the ability to gather qualitative and quantitative data in an easy way, making sure that the data I collected is reliable as I choose whom to send the survey to, and it offers an easy analysis process due to the simple structure of the survey.

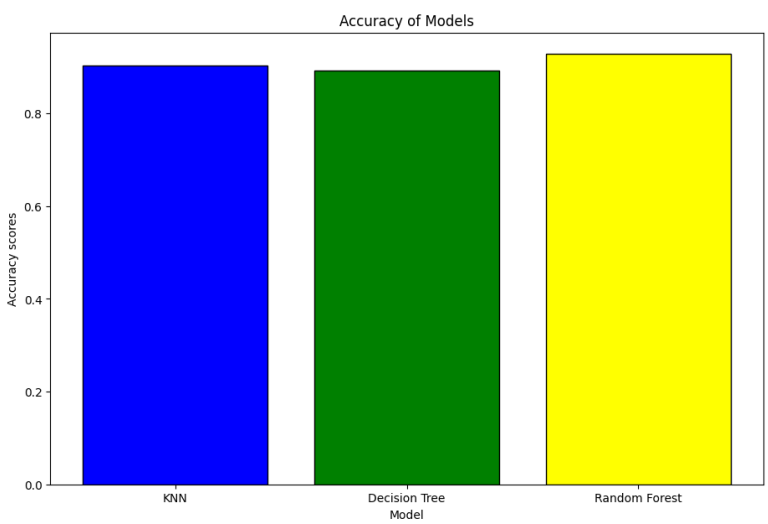
Strengths and weakness of the primary data:

The primary data helped to outline the ethical concerns, accuracy concerns, legal laws to put into consideration, benefits of the suggested AI and big data analytics system, and the challenges associated with implementing the system in Jordan, where the insights were directly collected by specialists making the data trustworthy and reliable. In addition, the data was tailored to answer the research questions and help achieve the research objectives and provided valuable insights as the survey included open-answer questions and provided qualitative data. On the other hand, there were limits to my primary data as the sample size was small which could not be inclusive of all the possible concerns, challenges, benefits, and impact.

3.2 Secondary data:

My choice of secondary data was a dataset from Kaggle that has the data of 310 patients who might have lung cancer, as it includes demographic information such as gender and age, daily habits such as smoking and alcohol consumption, symptoms such as yellow fingers, wheezing, coughing, shortness of breath, swallowing difficulty and chest pain, and other health factors that could indicate the presence of lung cancer such as anxiety, peer_pressure, chronic disease, fatigue, and allergy, and if the patient has lung cancer or not. I have used this data to test multiple machine-learning models to predict lung cancer, in order to suggest a machine-learning model that could be utilized in Jordanian hospitals.
Dataset link: https://www.kaggle.com/datasets/jillanisofttech/lung-cancer-detection

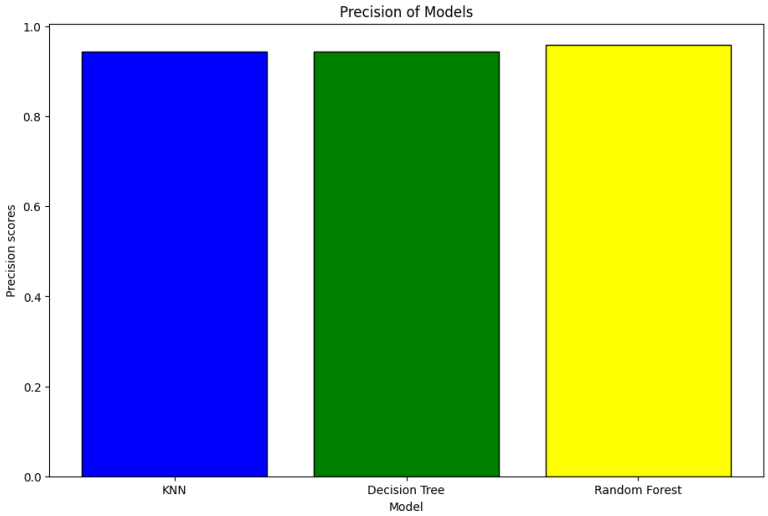
I have trained and tested 3 machine-learning algorithms: k-nearest neighbors (KNN), Decision Tree, and Random forest.




```
print(f'The Accuracy of KNN: {knn_accuracy}')
print(f'The Accuracy of Decision Tree: {dt_accuracy}')
print(f'The Accuracy of Random Forest: {rf_accuracy}')

The Accuracy of KNN: 0.9036144578313253
The Accuracy of Decision Tree: 0.891566265060241
The Accuracy of Random Forest: 0.927710843373494
```

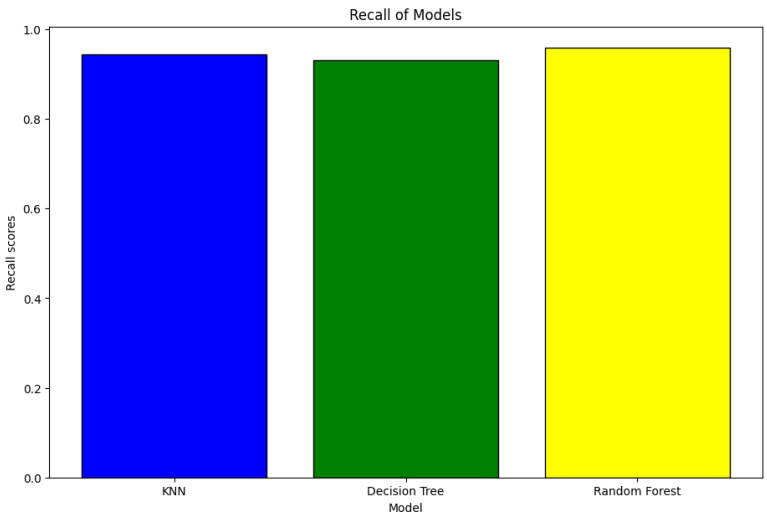
The results show that the Randon forest model had the highest accuracy which is the measure of how many times the model's prediction was correct.




```
print(f'The precision of KNN: {knn_precision}')
print(f'The precision of Decision Tree: {dt_precision}')
print(f'The precision of Random Forest: {rf_precision}')
```

```
The precision of KNN: 0.9444444444444444
The precision of Decision Tree: 0.9436619718309859
The precision of Random Forest: 0.9583333333333334
```

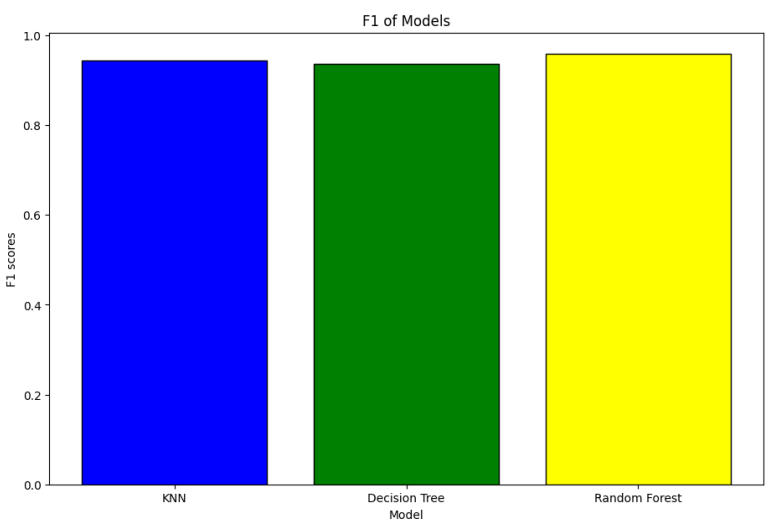
The results show that the Randon forest model had the highest precision score which is the measure of how many times the model's prediction of the positive cases was correct. In other words, how many times did the model predict that the patient had lung cancer and its prediction was correct?




```
print(f'The recall of KNN: {knn_recall}')
print(f'The recall of Decision Tree: {dt_recall}')
print(f'The recall of Random Forest: {rf_recall}')
```

```
The recall of KNN: 0.9444444444444444
The recall of Decision Tree: 0.9305555555555556
The recall of Random Forest: 0.9583333333333334
```

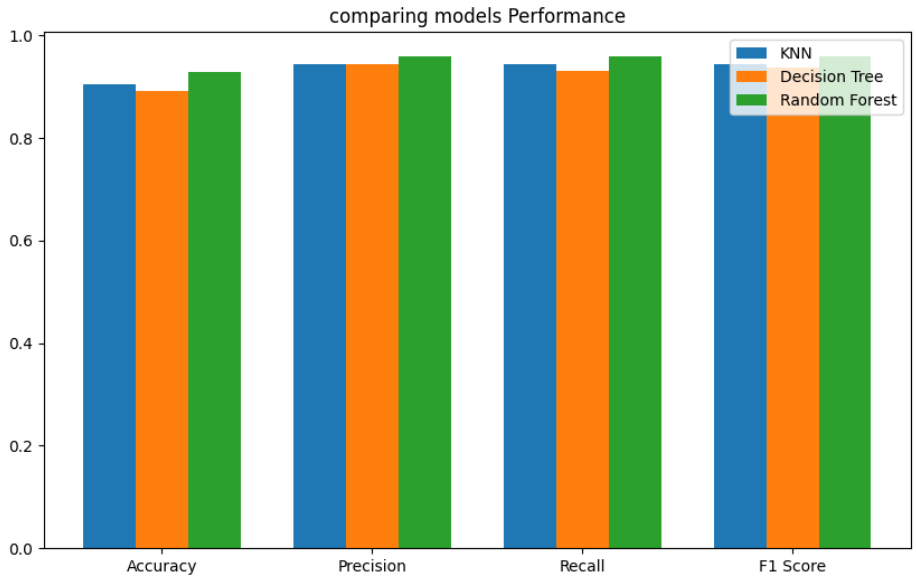
The results show that the Random forest model had the highest recall score which is the measure of how many times the model predicted true positive cases correctly.




```
print(f'The F1 score KNN: {knn_f1}')
print(f'The F1 score Decision Tree: {dt_f1}')
print(f'The F1 score Random Forest: {rf_f1}')
```

```
The F1 score KNN: 0.9444444444444444
The F1 score Decision Tree: 0.9370629370629372
The F1 score Random Forest: 0.9583333333333334
```

The results show that the Randon forest model had the highest F1 score which is the harmonic average of the recall and precision scores.



When comparing the overall performance of the models we can indicate that the Random forest model had the best performance as it has the highest Accuracy, precision, and F1 score making it the most suitable model to be used for early predicting lung cancer.

Justification for choosing this dataset as my secondary data:

This dataset is suitable to be used in my research because it is in alignment with my research topic and it would provide a baseline that could be further refined of what is the appropriate machine learning model to be used to early predict lung cancer by using it as the training data which is a part of my research questions and objectives. In addition, it includes realistic and a variety of potential lung cancer patients' information that could indicate the presence of lung cancer such as demographic information, symptoms, daily habits, and other health factors.

Strengths and weakness of the secondary data:

The dataset chosen covers multiple aspects of the patient's profile, enabling the model to predict lung cancer depending on a diversity of risk factors that indicate lung cancer. In addition, the dataset is structured and suitable to use with machine learning algorithms and doesn't require many preprocessing steps. On the other hand, the data has some drawbacks such as not being comprehensive of all the lung cancer indicators where lung cancer symptoms and risk indicators are not only limited to the attributes in the dataset, and the sample size is limited where 310 samples are not enough to test the model as in real life healthcare institutions the data volume is significant and the data is more complex.

4. Research Approach and Methodologies

In this section, the overall research methodology used to conduct the research will be explained. The Onion model will comprehensively give an overview of how the research was made starting from the general philosophy down to methods used for data analysis and collection. In addition, a flow chart of how the study was completed will be provided with an explanation of each step.

4.1 Onion Model

The "onion model" is a tool that helps the researchers conduct effective and efficient research, as it starts with the bigger picture of the research philosophy and guides the researcher step by step until narrowing down to the specific method used for collecting and analyzing data. This model helps ensure that the process of answering the research questions and objectives is well-thought-out and effective.

4.1.1 Philosophy

This layer focuses on the underlying hypothesis or beliefs of the research.
Methodologies:
1. Positivism: this method is used when the research belief is an undeniable fact.
2. Critical Realism: this method is used when the hypothesis is an undeniable fact, but people's experience might affect how they see this fact.
3. Interpretivism: this method is used when the research hypothesis relies on people's opinions and experiences.
4. post-modernism: this method focuses on thinking out of the box, where it takes an untraditional approach and tries to challenge the known.
5. pragmatism: this method believes that experiments are the most effective way to prove a hypothesis.

In my research, I have chosen the Interpretivism method, where my research topic highly relies on people's opinions, as each oncology doctor, data analyst, or expert in my area of research could have different opinions regarding the ethical concerns, challenges, impact, and benefits of utilizing the suggested early lung cancer prediction system in Jordan.

4.1.2 Theory Development Approach

1. Deduction: this approach is used when there is a clear theory or principle from the beginning, then it is tested and experiments are done to reach an answer to the research.
2. Induction: this approach starts with an observation, then it is developed until reaching a clear principle or theory.
3. Abduction: this approach is similar to the induction approach. Where it starts with the observation but it doesn't reach a clear and certain principle or theory.

In my research, I have chosen the deduction approach where there were research papers already discussing lung cancer prediction using AI and big data analytics thus the basic principle and theory already exist. I started reviewing these papers then conducted a survey and did experiments inspired by the information that the previous research papers provided helping me address my research questions and objectives, due to that I started by reviewing previous

research papers and then testing the theory, my theory development approach is classified as deduction.

4.1.3 Methodological Choice

1. Mono method quantitative: in this method, the nature of the data used to conduct the research is numerical or quantified.
2. Mono method qualitative: in this method, the nature of the data used to conduct the research is non-numerical or includes a description.
3. Multimethod quantitative: in this method, numerical and non-numerical data are used to conduct the research but the numerical data is more focused on.
4. Multimethod qualitative: in this method, numerical and non-numerical data are used to conduct the research but the non-numerical data is more focused on.
5. Mixed method simple: in this method, quantitative and qualitative data are used equally to conduct the research, where one or two research strategies are used.
6. Mixed method complex: in this method, quantitative and qualitative data are used equally to conduct the research, where multiple research strategies are used.

In my research, I have used the mixed method simple, where I conducted a survey that has qualitative and quantitative questions in order to gather the specialists' responses, I included qualitative questions to enable the specialists to express their concerns and opinions without limiting them with specific answers, ensuring that the data collected through the survey is comprehensive. In addition, I have used an experimental research strategy to test multiple machine learning models to suggest a suitable model for early predicting lung cancer in Jordanian hospitals. I have chosen this method because it enables me to gather valuable insights as combining qualitative data that provide in-depth insights and quantitative data that provide statistical insights will help provide comprehensive and effective research.

4.1.4 Research Strategy

1. Experiment: this strategy focuses on experience where it gathers data by testing multiple approaches or methods.
2. Survey: in this strategy, data is collected through questionaries made by the researcher and sent to the population he is focusing on.
3. Case study: in this strategy, data is collected by studying and exploring issues within real-life contexts.
4. Archival study: in this strategy, data is collected by reviewing previous research papers and documents related to the research topic.
5. Ethnography: in this strategy, data is collected by observation.
6. Action research: in this strategy, data is collected by experimenting with different strategies and adjusting them.
7. Narrative inquiry: in this strategy, data is collected by conducting interviews with experts in the research field.
8. Grounded theory: the grounded theory takes two different samples and compares between them, and it is a qualitative strategy.

In my research, I have conducted a survey, where it is easier and more flexible for the specialists to fill it out whenever they are free, and it is crucial in my research to gather the data directly

from the experts where they have a comprehensive understanding of what are the legal laws to be considered, legal concerns, potential benefits, and the impact on patients associated with the early lung cancer prediction system. In addition, I have used the experiment research strategy, as it is vital to test multiple machine learning models and evaluate their accuracy before suggesting a model to be used in the system.

4.1.5 Time Horizons

1. Cross-sectional: in this method, the research is not affected by time.
2. longitudinal: in this method, the research is affected by time.

I have used the cross-sectional in my research where my research objectives and questions are not affected by time.

4.1.6 Techniques and Procedures

In order to extract valuable insights from my survey and secondary data I have started by analyzing the survey responses, where Google Forms provides analysis charts that I have used to identify and highlight what the majority of the specialists think regarding the ethical concerns, challenges, benefits, impact on patients, patients data privacy concerns and other aspects to pay attention to when integrating big data analytics and AI for early lung cancer prediction in Jordanian hospitals. This has helped me answer my research questions and objectives as I was able to get the specialists' opinions and points of view and analyze them effectively.

To improve my research, I have used a sample dataset from Kaggle that has information related to potential lung cancer patients to test multiple machine learning models, where I have trained and tested the Decision tree, KNN, and Random forest model with the help of Scikit learn library. To analyze and identify which model has the best performance I have calculated each model's accuracy, recall, f1, and precision and used Matplotlib library to visualize the results such as using bar plots to plot the evaluation matrices scores to have a better overview of which model is the most suitable to be used in the early lung cancer prediction system.

4.2 Research Methodology

Justification and evaluation:

1. Literature review:
In this step previous research papers were reviewed, starting from research papers discussing The difficulty of early diagnosing lung cancer and its importance, moving to a research paper reviewing the benefits and concerns associated with utilizing big data in healthcare, digging deeper into utilizing AI and big data in cancer diagnosis, and finally narrowing down to research papers more specific discussing AI and big data approaches in lung cancer diagnosis. in my literature review I have used both Thematic and Chronological.

This step is crucial as by reviewing previous research papers I was able to establish a better understanding of what are the gaps that I should try to fill in my research in order to ensure that

my research is valuable and beneficial. In addition, it helped me set clear and efficient research objectives and questions.
Evaluation:
The literature review conducted is comprehensive as it covers multiple aspects such as early lung cancer diagnosis challenges, big data in healthcare, concerns and potential of utilizing AI and big data in cancer diagnosis, and AI and bid data specifically in lung cancer diagnosis. In addition, it is well-structured where the papers were ordered Thematicly which gives a comprehensive overview of different aspects, and the research papers specifically discussing my topic were ordered depending on the date of publishing to give an overview of how these systems have developed and what are the currently suggested algorithms which will help conduct a comprehensive and up-to-date research.

2. Setting clear objectives and research questions:
In this step, the aim and the path that my research will take will be defined. This step is vital, and the research objectives and questions should be set after good consideration where I have made sure that it will be targeted to the Jordanian population and addresses the gaps defined in the literature review, as it determines how efficient and effective the research is, and without setting clear research objectives and questions the research would be un-structured.

Evaluation:
The research questions and objectives of my study were set precisely, where they are not generalized and clear. Making my research effective and well-structured. On the other hand, my research's scope is limited where there are aspects of utilizing AI and big data for early lung cancer prediction systems in Jordan that weren't covered in my research objectives and questions due to not having a long period of time and limited resources.

3. Conducting a survey:
My choice of primary data was conducting a survey that will help me gather qualitative and quantitative data. Conducting a survey helped me answer my research questions and achieve my research objective by getting responses directly from doctors and experts in the data science field. This step is crucial as collecting primary data is one of the main elements of effective and efficient research.

Evaluation:
My choice of collecting primary data through surveys was highly effective, as it helped me gather valuable real-world insights from the specialists' and doctors' experience, another benefit gained from choosing the survey research strategy is that it has freed me and the specialists from time constraints where they can fill it whenever they are free and it will be less time consuming to me as I only created the survey and sent it. In addition, including qualitative questions helped gather more valuable insights as the specialists could express themselves without limiting them to specific answers, and the quantitative data provided valuable statistical insights. However, there is a drawback of my primary data where the sample size is small.

4. Analyzing the Survey Responses:
After the survey was filled out by the specialists, it is crucial to analyze the results where the survey and primary data gathered become useless without highlighting what it represents and

indicates. I used the charts provided by Google Forms as it helps have a better understanding of the most repeated answers such as what are the ethical concerns associated with implementing an early lung cancer diagnosis system using AI and big data that the majority of the specialists agree on which provides an answer for one of the research questions.

Evaluation:
My choice of analysis tool is charts provided by Google Forms and underlining what are the most repeated answers to the qualitative questions was effective and efficient as I could easily highlight what the specialists agreed on. In addition, Google Forms charts are clear and easy to understand making them usable in the research report.

5. Pre-processing the Dataset:
The pre-processing step is crucial where the accuracy of the models depends on the quality of data, as null values should be filled, certain features should be normalized if there was deference in the values where certain features have much higher values than the other ones, and categorical values should be turned into numerical values so they can be fed to the model. In order to address these obstacles normalization, filling null values and encoding should be done.
In my case, my dataset didn't need much pre-processing where I only needed to encode categorical values, normalize certain features, and drop duplicated rows.

Evaluation:
The pre-processing step was vital in my research where it helped me raise the efficiency and effectiveness of my model's performance as higher quality data was used. On the other hand, the dataset used is small which may not be similar to the real-life datasets, so there will be a need for much more pre-processing due to the complex nature of the medical data implemented in real life.

6. Training and Testing Multiple Machine Learning Models:
In this step, I have tested multiple classification models from the Scikit learn library such as KNN, Random forest, and Decision tree. As testing multiple models helps explore and compare the options and reach the best accuracy which is crucial when predicting lung cancer.

Evaluation:
My approach of testing multiple models was highly effective as I was able to find the convenient model for my research which is Random forest that had high accuracy (0.927), which meets the need for high accuracy when diagnosing lung cancer. On the other hand, the dataset used for the prediction is small which may not be similar to the real-life datasets, so the model will need to be refined when implemented in real life.

7. Evaluating Each Model's Performance:
In this step, the Accuracy, precision, f1, and recall evaluation matrices were calculated for each model, in order to compare which model had the best performance and highlight the strengths and weaknesses of each model.

Evaluation:

Evaluating each model's performance was crucial, as it provides a better understanding of which model is better to be used in the system and makes the decision more informed and efficient. In addition, my choice of the evaluation matrices was effective as it gives a comprehensive overview of the false positive, true positive, false negative, and true negative predictions.

8. Analyzing and Comparing the Models' Performance:
In this step, the models' performance is compared to decide which model is the best to be used in early lung cancer diagnosis systems. Without this step, there will not be any use
In order to analyze the results and make a more informed decision I have visualized the results using Matplotlib library.

Evaluation:
This step was crucial and was highly effective, as using Matplotlib to visualize the results helped add a better overview of the models' performance and easily compare the performance of the models resulting in making a more informed decision.

5. Results and Discussion

From the survey conducted and the experience of multiple models, I was able to get valuable insights that would help me answer my research questions and objectives.
Starting from the survey conducted that was sent to doctors and data science specialists.

I was able to indicate that:

From your perspective, what are the ethical concerns that should be taken into consideration when integrating this system into Jordanian hospitals?

15 responses

- patient consent
- the patient's nonacceptance of being diagnosed by AI
- the need for taking the patient's consent
- Patient concent- for the doctor not to be biased with his diagnosis because of the results on the AI technology diagnosis, but to integrate both his medical opinion and the AI technology.
- patient consent and trust with the AI algorithm
- patient data privacy
  Bias in AI algorithms
  Consent for AI-based diagnosis
  Transparency in AI decision-making
- Patients should be informed that diagnosis is done with the help of AI.

From your perspective, what are the ethical concerns that should be taken into consideration when integrating this system into Jordanian hospitals?

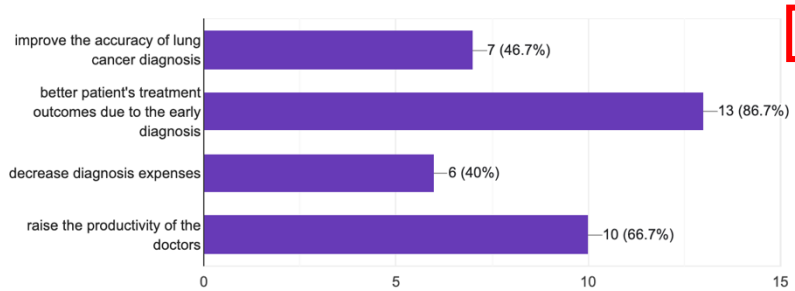
15 responses

- Patients should be informed that diagnosis is done with the help of AI.
- the patient not approving for being diagnosed using machines and not human
- Consent of the patient
- being transparent with the patients about how their data will be used and how they will be diagnosed
- the transparency of how the decision was made
- the doctor's bias toward the AI prediction without making sure of the diagnosis
- culture acceptance of being diagnosed by AI
- patient consent and being transparent with the patient about how the decision was made
- Patient right to informed consent, the use of AI as a diagnostic tool instead of being used to assist with diagnosis.

- The major ethical concerns to be considered when implementing a system that utilizes big data and AI for early lung cancer diagnosis in Jordan are obtaining the patients' consent and acceptance that their diagnosis was made with the help of AI, and the concern of the doctors being bias to the AI system's diagnosis and not utilizing his expertise.

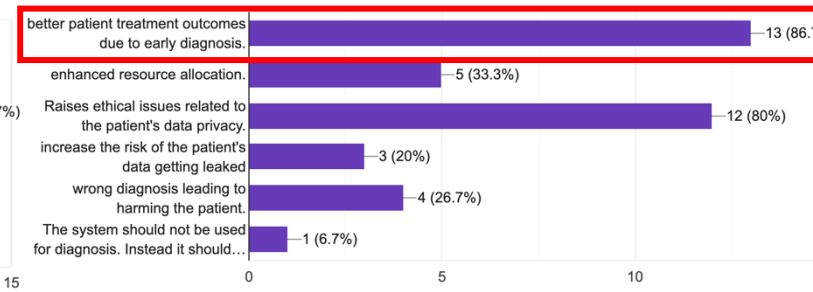
From your perspective, how would integrating AI and big data analytics for early lung cancer diagnoses benefit the Jordanian healthcare industry?

15 responses



From your point of view, how would utilizing AI and big data analytics (the new system) in e cancer diagnosis impact the patients?

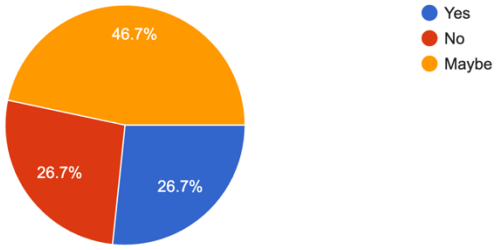
15 responses



- The major benefits of utilizing these systems in Jordanian hospitals are improving the patients' treatment outcomes due to early diagnosis and raising the doctor's productivity.

Do you think that changing the currently used framework will cause an obstacle?

15 responses



if yes please justify your answer

4 responses

It would take a long time transfering data and might lead to obstructing the work of doctors temporarily .

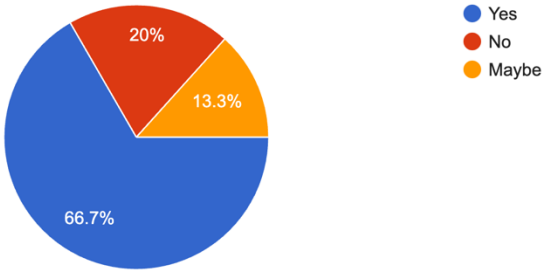
the need for a robust mechanism for transferring the patient data to the new system

losing the patient's data

The framework took years of work and a lot of resources to develop and upending the whole system will be extremely costly

Do you have concerns regarding the patient's privacy and data security?

15 responses



If yes from your point of view how can we address them?

10 responses

cybersecurity regulations and encryption

patient data breach will cause legal issues

Setting a very strict regulations on the staff to insure protecting it- setting punishments for leaking any data such as salary deduction .

encrypting the patients data and only giving data access privilege to the individuals that need to

Selling patients data to third parties for commercial reasons including insurance companies.

strict regulations and Legal liability to ensure the patient's data privacyt

by ensuring the presence of a powerful security system

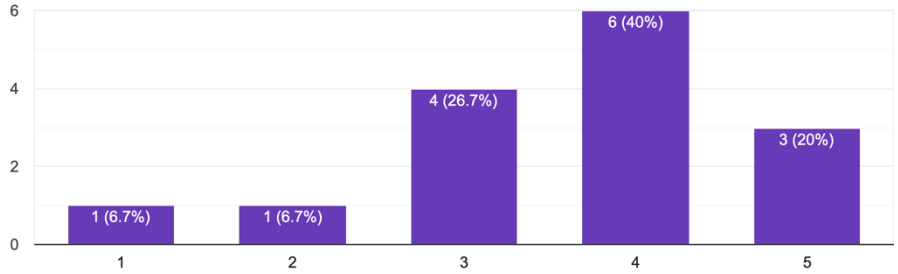
implementing a strict protection mechanism

the first priority that should be considered before implementing the system is establishing a powerful and strict data protection regulations

To ask for patient consent and make sure the consent is informed. The use of security measures to encrypt and protect patient data

Do you have concerns regarding the accuracy of the system prediction?

15 responses



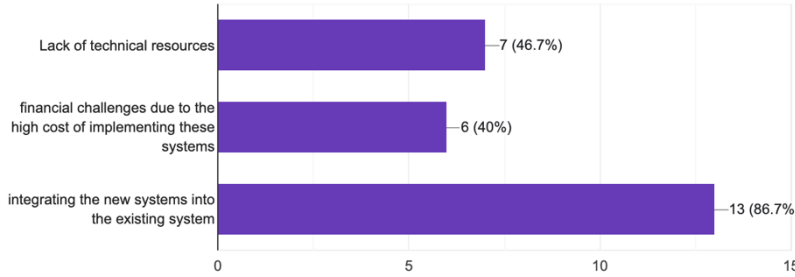
please mention the concerns that you have regarding the accuracy of the system

15 responses

- False negative will cause the patients to relax while the disease is progressing.
- The system should not be used to diagnose patients but to asses risk of developing lung cancer. Diagnoses without the use of AI with years of human experience and knowledge are still inaccurate in some cases and AI is too underdeveloped to be used in this manner
- if the patient was misdiagnosed with lung cancer and was harmed due to the unnecessary chemotherapy
- if the patient wasn't diagnosed correctly the doctor and the hospital will be legally harmed
- diagnosis must be reviewed by the doctors because any mistake will affect the doctors reputation significantly
- false negative and false positive
- the dr depending on the AI system, that could be not accurate in some cases resulting in harming and miss diagnosing the patient
- false positive and false negative
- AI hasn't reached the point where it could be completely reliable and it could miss diagnose patients which is not acceptable

What are the challenges that could be faced when implementing these systems in Jordanian hospitals?

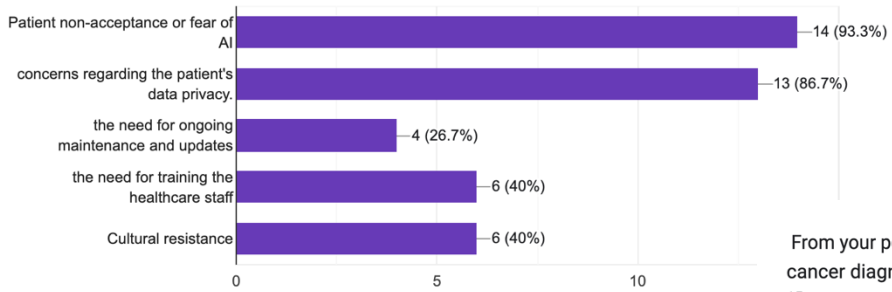
15 responses



- The main challenges and concerns that should be focused on when implementing the early diagnosis system in Jordan are integrating the new system into the existing system as there are concerns regarding transferring the data, losing the patient data, and expenses. In addition, patient data privacy Is a major concern, that should be addressed by implementing a powerful patient data security system, using encryption tools, legal liability, and strict regulations to be set for the staff. Another concern that was discussed is the accuracy of the system's diagnosis as there are concerns regarding the false positive and false negative where if the patient was miss-diagnosed with lung cancer he could be harmed due to the unnecessary chemotherapy, on the other hand, if he was miss-diagnosed as lung cancer free his cancer could develop to advanced stages while he needed to be treated, and AI still has not reached the point where it can be fully accurate and completely relied on and the concern of miss diagnosis by AI could cause legal liability and harm the reputation of the doctors and the hospital.

From your point of view what are the obstacles that could be faced after implementing this system related to adapting to the new system?

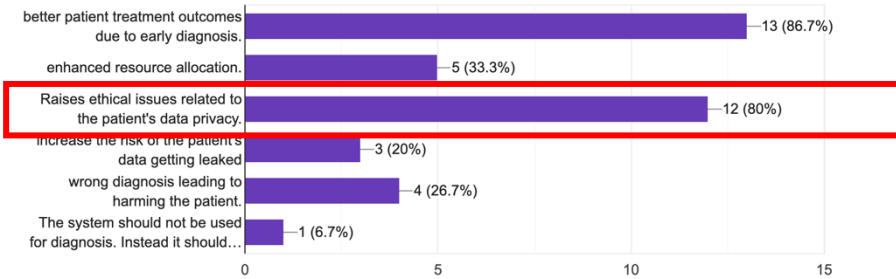
15 responses



- The main challenges that could be faced after implementing the system in Jordan are the patient non-acceptance or fear of AI, and concerns regarding patient data privacy.

From your point of view, how would utilizing AI and big data analytics (the new system) in early lung cancer diagnosis impact the patients?

15 responses



what are the technologies currently used for diagnosing lung cancer?

15 responses

- Imaging techniques
- Medications taken by the patient, blood tests, and imaging technologies
- CT-scans and blood test
- Radiology - Ct scan- MRI- tumor marker- biopsy- PET position emission tomography- lung function tedt
- imaging technologies and lung function test
- CT scans PET scans Chest X-rays Biopsies Blood tests
- -
- PET, tumor marker, blood test, and biopsy

what are the technologies currently used for diagnosing lung cancer?

15 responses

- PET, tumor marker, blood test, and biopsy
- Tumor markers , CT , MRI , Biopsy , PET
- reviewing the electronic medical records to know the patient's medical history
- EMR
- biopsy , ct scan , MRI, lung function test and EMR to review the diseases the patient had and the medications he was on
- CT scan, X-ray and biopsy
- biopsy and patients medical history
- Imaging technology used in diagnosis of all types of cancer as well as blood and biopsy testing and imaging technologies used for all cancer types

- The technologies currently used for diagnosing lung cancer are imaging techniques, blood tests, biopsies, EMR, and the patient's medical history including medications they took and diseases they were diagnosed with.

What are the Jordanian legal laws that should be taken into consideration before implementing these systems?

15 responses

- MInistry of health patient protection laws
- Doctor and Patient confidentiality
- electronic data privacy regulations
- Protecting the patient privacy
- Minister of Health approval and alignment with its regulations
- Data protection laws Medical liability regulations AI and machine learning governance Patient consent laws
- Data privacy laws.
- ministry of health approval of the system

What are the Jordanian legal laws that should be taken into consideration before implementing these systems?

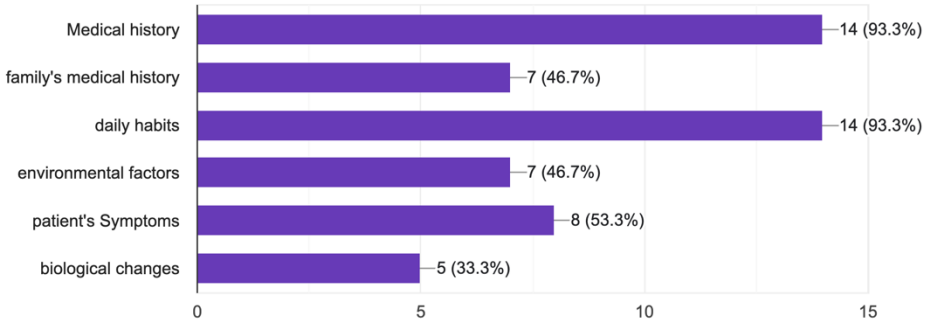
15 responses

- Data privacy laws.
- ministry of health approval of the system
- Insuraning patient privacy
- minister of health patients protection laws
- doctor patient confidentiality
- not my area of expertise
- minister of health and Jordan Food and Drug Administration laws
- - -
- Not my field of expertise

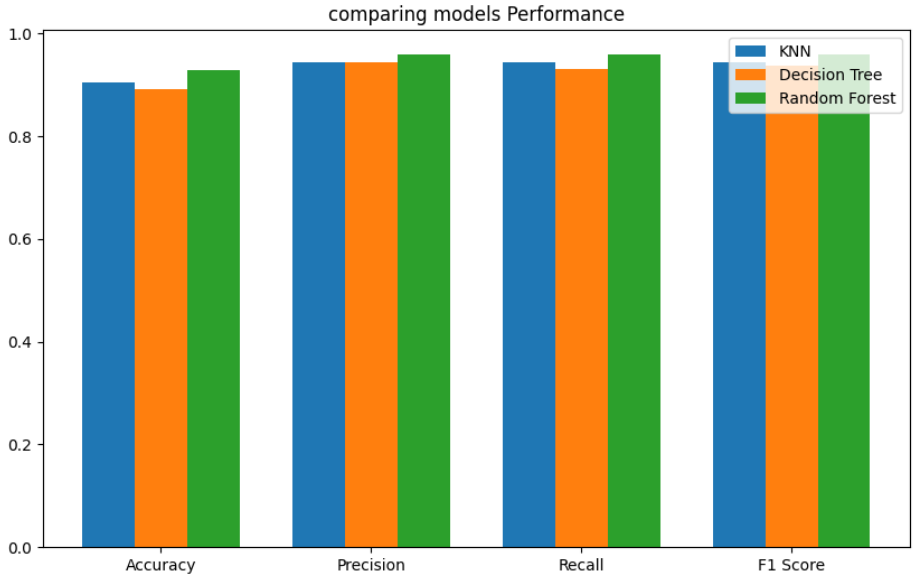
- One of the significant aspects to pay attention to when implementing the early lung cancer diagnosis system in Jordan is the legal laws that should be followed such as the Ministry of Health's legal laws, data privacy, patient protection, and patient-doctor confidentiality.

What are the criteria that should be followed in order to diagnose the patients?

15 responses



- The criteria that should be followed to diagnose the patients is their daily habits and medical records



- As for the experiment which was testing multiple models I found that the Random forest model was the most convenient to use as its performance was the best compared to the Decision tree and KNN.

How the results meet the research question and objectives?

By analysing the data gathered from the survey I was able to answer my research questions and objectives.

- What would the early lung cancer diagnosis systems offer the Jordanian healthcare systems? By gathering insights into what the doctors and data specialists in Jordan think the benefits of integrating such systems in Jordan are I was able to answer this question.
- What are the challenges that come along with integrating AI and big data analytics systems into the Jordanian healthcare sector? After analyzing I was able to identify major challenges as different questions were asked covering different aspects so I was able to comprehensively answer my question.
- What are the ethical and legal concerns that should be considered when implementing these systems in Jordan? was answered by gathering insights from the doctors' and specialists' responses about whet they think the ethical concerns are and other questions related to the ethical concerns.
- What is the suitable machine learning algorithm to be used in early lung cancer prediction? I was able to answer this.

As for the research objectives:

- I was able to highlight the Jordanian community's ethical concerns and Jordanian laws that should be considered when implementing the system by analyzing the responses of the survey where the responses of the doctors and the specialists helped underline the ethical concerns and the legal laws to be considered and addressed before implementing the system.
- To establish a basic guideline that could be further refined on how these systems would be implemented and utilized to help ease and make the diagnosis process more effective resulting in decreasing the number of deaths caused by lung cancer in Jordan. I was able to achieve this objective as through my research I highlighted what are the steps that should be taken to implement the system starting from what are the legal laws that should be considered, how to protect the patient data which is crucial, the need for obtaining the patient's consent, reaching what is the convenient model to use. Resulting in raising the possibility of actually implementing these systems in Jordan aiming to help make the

diagnosis process more effective and decrease the number of deaths caused by lung cancer in Jordan.
Evaluate the Potential Impact and Challenges associated with implementing an AI and big data-based early lung cancer diagnosis system in Jordan. I was able to achieve these objectives as the participants were asked about the impact and challenges in a direct and indirect way which was demonstrated and highlighted in the results.

The merits and drawbacks of the analysis:

The survey analysis had its strengths and weaknesses. As the insights were directly collected from healthcare professionals and data science specialists including qualitative and quantitative data which provided valuable and reliable insights. In addition, the analysis helped me comprehensively answer and achieve my research questions and objectives, as the answer to each research question was reached by analyzing multiple questions covering multiple aspects. On the other hand, the sample size of the data was small resulting in not making the analysis highly reliable and comprehensive. In addition, some questions gave surface-level Insights which may not provide detailed analysis or detailed explanations. As for the model's performance analysis, it was effective as the model's performance was evaluated depending on multiple matrices each measuring an aspect resulting in making a more informed decision and it has given me an actionable recommendation as it was clear that the Random forest model had the best performance. However, the experiment analysis was not an in-depth analysis due to the black box nature of the algorithms resulting in not providing enough details regarding how the model works.

6. Conclusion and Recommendations

The aim of this research was to explore and investigate the challenges and benefits of integrating AI and big data based AI and lung cancer diagnosis systems in Jordan. Seeking to especially highlight the impact on the Jordanian healthcare system, identifying integration challenges, considering Jordanian ethical and legal issues, and recommending suitable machine learning models aiming to provide a baseline of how the system could be implemented in Jordan to be able to raise the efficiency of lung cancer diagnosis in Jordanian hospitals.

I was able to answer the research questions and objectives and achieve the research aim through the findings summarized as:

- The ethical concerns to be considered when implementing the system in Jordan are: obtaining the patients' consent and acceptance that their diagnosis was made with the help of AI, and the concern of the doctors being biased to the AI system's diagnosis and not utilizing his expertise.
- The benefits of implementing the system in Jordan hospitals: are improving the patients' treatment outcome due to the early diagnosis and raising the doctor's productivity.
- The major challenges to be considered when implementing the system in Jordan are integrating the new system into the existing system, patient data privacy that should be addressed, and the accuracy of the system's diagnosis.
- The major challenges that could be faced after implementing the system in Jordan are the patient non-acceptance or fear of AI, and concerns regarding patient data privacy.

- The legal laws to be considered when implementing the system in Jordan are the Ministry of Health's legal laws, data privacy, patient protection, and patient-doctor confidentiality.
- The technologies currently used for diagnosing lung cancer in Jordan: are imaging techniques, blood tests, biopsies, EMR, and the patient's medical history including medications they took and diseases they were diagnosed with.
- The criteria that should be followed to diagnose the patients in Jordan are patients' daily habits and medical records.
- The suitable model to be used when implementing the system is Random Forest.

Recommendations:

Based on the findings in order to implement the early lung cancer diagnosis system using AI and big data in Jordan I recommend:

1. sharing awareness of how the patient's data will be used by the system and ensuring the Jordanian patients that their data is protected and used ethically and without causing any harm in order to make them feel comfortable giving their consent and feel more comfortable with the idea of being diagnosed by AI and not being intimidated by it.

2. The system should be tested for a while and further trained and the doctors should not fully depend on the system's diagnosis but also utilize their expertise to make sure of the diagnosis.

3. A powerful data protection mechanism should be set and security systems must be implemented before utilizing the early lung cancer diagnosis system.

4. Consult with a legal consultant about what are the legal regulations that should be taken to be in alignment with the Ministry of Health's legal laws, data privacy, patient protection, and patient-doctor confidentiality laws.

5. I suggest using the Random forest machine learning model when implementing AI and big data based early lung cancer diagnosis systems.

Future work:

- There is a need for further research on how to integrate the early lung cancer prediction systems into the Jordanian hospitals' existing systems without facing obstacles in transferring patient data that has a huge volume or needing to completely change the existing system which will cause lossing this data which would be a significant problem.
- Refining the system and improving it where it could use imaging data to diagnose patients as well.

7. Reflections

7.1 Selected Research Methodology

When starting with the research the process seemed overwhelming and complicated where many details were involved. In order to ease the process and ensure that I covered all the aspects that make my research valuable and comprehensive I started brainstorming to decide what is the

convenient methodology to be used in my research and took it from there step by step until the research was completed.

- **Literature Review**

In order to establish a better understanding of what path I should take and what area of early lung cancer diagnosis systems I should focus on, I started reviewing previous research papers in order to highlight the gaps and areas of improvement. In the beginning, I struggled a lot in this step and felt overwhelmed as there were many aspects and details. I started by only reviewing research papers that specifically discussed lung cancer prediction systems using AI and big data and mainly focused on suggesting algorithms, this has caused an obstacle where there wasn't an explanation or focus on basic guidelines or enough focus on aspects other than the suggested approaches which has caused information overload and made the picture unclear, where I wasn't able to highlight what area I should focus on due to starting with advanced research papers. To overcome this obstacle I started simplifying the process by reviewing generalized papers that discussed the importance of early lung cancer diagnosis, the opportunities and challenges of integrating big data in healthcare, opportunities and challenges of integrating AI and big data for cancer diagnosis, then started focusing on the research papers specifically discussing my research topic. This has given me a more comprehensive understanding of the research area and what I should focus on. In addition, one of the difficulties I have run into is determining which research papers I should use as there was a wide range of options and it was hard to decide which ones are the most suitable and fits my area of research. After finishing the Literature Review I felt satisfied and proud as this was a crucial step and at the beginning I faced many challenges but I was able to overcome them. This step was well thought out and I made so many adjustments until I was satisfied with the final output, but if I had more time I would have added more research Papers that focus on the opportunities and challenges of using AI and big data for early cancer prediction and discuss the ethical concerns, legal laws, and technical challenges in depth.

Merits:

Completing the Literature review helped me establish a crucial and foundational understanding of my research topic and highlight the gaps that should be filled, ensuring that my research is valuable and contributes new insights.

Limitations:

As valuable as the literature review was, it was time-consuming to search for suitable research papers, and summarizing them takes a lot of time. In addition, the general research papers that discussed big data in healthcare and lung cancer early diagnosis importance were not up to date compared to the others where there are no recent papers that focus on these topics.

Pitfalls:

The literature review could limit innovation as there might be bias to the opinions and limitations of the previous paper. In addition, especially when it comes to AI and big data there is a rapid evolution, resulting in not being able to stay on track with the enormous amount of new research papers which may make my research not up to date quickly.

- **Setting Clear Research Objectives and Questions**

Setting the research objectives and questions was a crucial step as the research path depended on it. Setting the objectives and questions was much easier and made me feel relieved as the path of my research started to get clearer because my literature review was comprehensive and helped me highlight the gaps effectively which resulted in setting efficient and straightforward research objectives and questions. I have set the research questions and objectives carefully and it was comprehensive of the crucial aspects. However, if I had more time I would make my research deeper and make my scope larger where it doesn't only focus on the crucial and basic aspects but focuses on the complementary aspects as well.
Merits:
The merits of my research questions and objectives are that it was straight to the point and it was comprehensive as they covered the crucial aspects that should be paid attention to when implementing such systems in Jordan. In addition, it helped make the research focused and structured.

Limitations:
The objectives and questions might limit the scope of exploration and reveal insights that were not considered, as the focus will only be on addressing the research questions and objectives.

Pitfalls:
If the research questions were not set accurately and were not comprehensive it will directly affect the effectiveness and value of the research.

- **Conducting a Survey**

When choosing the research strategy, I struggled to choose the suitable one as there were many options, to overcome this struggle I started by evaluating what each strategy would offer my research and decided that the survey was the most suitable as I would be able to directly gather insights from the experts in my area of research and it frees the experts and me from time limits where they can fill it any time. Another struggle was reaching the experts as due to the nature of their jobs they didn't have enough time to respond to my messages and fill out the survey, which resulted in not being able to gather a decent amount of responses and made my sample size small. Although my survey was conducted effectively It would be much more effective if I was able to gather more responses, interview an oncology doctor, and conduct another survey that will be sent to Jordanian lung cancer patients to evaluate how they will react to being diagnosed by the help of AI and how would they be more comfortable with it. This would ensure that my analysis is more comprehensive and reliable.
Merits:
The survey enabled me to gather reliable primary data that is specific and relevant to the research questions and objectives, as the data was directly gathered from the experts that I have chosen and the questions were written by me which I ensured that they were comprehensive and helped me effectively address my research questions and objectives. In addition, using the mixed method helped me gather valuable insights as the qualitative data helped me gather in-depth insights and the quantitative data helped me gather statistical insights.

Limitations:
The designing phase of the survey questions could be time-consuming and require a lot of attention and effort to ensure that they address my research questions. In addition, gathering a decent amount of responses can be difficult and in my case, I was not able to gather the number of responses I desired.

Pitfalls:
There is a potential for bias, where the doctors could be working in the same hospital or the experts might be working in the same company, as I informed the experts and doctors to feel free to send the survey to their workmates.

- **Analyzing the Survey Responses**

This was a step that should be carefully done as the efficiency of the research highly depends on it. I didn't face any challenges when analyzing the survey responses and I was confident of the approach I was taking, when creating and setting the survey questions I paid close attention to how each question would help to complete the research so I already had In mind how my analysis structure would be. When finishing this step the research got much clearer and the research questions and objectives were addressed. In addition, the plots provided by Google Forms made the analysis process much easier thus, I would not change anything in my survey analysis step.
Merits:
The plots provided by Google Forms were highly effective as they eased up the analysis process and it provided will structured and organized analysis structure which enabled me to layout the results in an organized way. In addition, my analysis was comprehensive as multiple aspects were covered and the research question and objectives were addressed. The method used was mixed where qualitative questions were included, enabling the participants to express their opinions which helps gathering more valuable insights.

Limitations:
The sample size was small which might affect the effectiveness of the analysis.

Pitfalls:
There could be a misunderstanding of the expert's and doctors' opinions which could affect the integrity of the research. However, in the qualitative questions analysis, I have focused on the most repeated question which helped avoid this problem.

- **Pre-processing the Dataset**

In most cases, the pre-processing step is time-consuming. However, my dataset was already pre-processed and didn't have any null values, so I only needed to normalize and encode certain features. This step was easy and I was able to finish it quickly as I already have experience in pre-processing. due to the simplicity of my code and the nature of the data set, there will be no need for any other pre-processing step or adjustments to be done on this step.
Merits:
The pre-processing steps improve the accuracy of the models, as machine learning models highly rely on the quality of the data.

Limitations:
There weren't any major limitations in the pre-processing step. However there might be limitations unconsidered as my pre-processing steps were quite simple which could make it un-comprehensive, such as there wasn't any outlier detection or checking for imbalanced data.

Pitfalls:
There is a possibility that overfitting or underfitting occurs if the data wasn't accurately pre-processed. However, my code's output doesn't indicate the presence of overfitting or underfitting.

- **Training and Testing Multiple Machine Learning Models**

I really enjoyed this step, as I have experience and passion in the machine learning field, so I was confident of my steps and didn't face any challenges or obstacles and the results were satisfying. If I would change anything in this step I would test more models such as SVM and neural networks models as it would give me more options to choose from.
Merits:
The models tested were commonly used and known for their effectiveness, and the best model was random Forest which had excellent accuracy(0.927) which indicates that it is suitable to be suggested as an algorithm to be used for early lung cancer prediction especially since it is significantly important that the diagnosis is made accurately.

Limitations:
When Integrating these systems in real life there might be a concern of needing robust computing power.

Pitfalls:
The models might not give high accuracy with all training datasets, making it crucial to find the reason behind this problem and how it could be solved.

- **Evaluating the models' performance**

This was one of the simplest steps in my research, as there are already built-in functions that calculate the evaluation matrices. When I completed this step I felt relieved as now the research questions and objective are completely addressed thus I would not change anything in this step.
Merits:
Evaluating each model's performance enables me to compare the models and decide which one is the most convenient one to use in my system while paying close attention to the fact that high accuracy is crucial when diagnosing lung cancer. In addition, all three models had good accuracy which gave me alternative options in case of future adjustments.

Limitations:
There were no major limitations in this step, but deciding which model is the best depending only on the performance, and ignoring the transparency of each model and how understandable the model is could be a limitation, especially in healthcare and lung cancer diagnosis.

Pitfalls:
Due to only using basic evaluation matrices, there could be aspects that need to be covered and I didn't pay attention to resulting in causing unseen obstacles. However, my code and the models used are quite simple and don't require advanced evaluation.

- **Analyzing and Comparing the Models' Performance**

This step was easy to do, as I already have experience with the library Matplotlib. I enjoyed this step as it was easy, didn't take much time, and made the results of the code much clearer. However, if I would change anything in the analysis step I would add different plots to cover more aspects.
Merits:
Analyzing the results using plots enables me to have a better overview of each model's performance and compare the overall performance of the models resulting in ensuring that the decision made is informed.

Limitations:
My analysis was not advanced as I only used bar plots due to the simple nature of my code and models. In addition, the analysis might not be understandable to non-expert in machine learning as they will not understand on what basis the model was chosen.

Pitfalls:
Relying on the quantitative aspects of the analysis and overlooking the qualitative aspects such as which model is easier to integrate into existing systems or transparency could cause an obstacle in the future.

- **Overall process reflection**

Although there were parts of this research that I have struggled with, I have enjoyed each step of the process, as I have learned how to conduct valuable research. At the beginning of the research, I felt overwhelmed as there were many aspects to cover, and many details, and my project aim was not clear and vague, however, step by step, and by finishing each phase of the research the picture got clearer and I became more confident. After finishing the research I felt so proud and satisfied with how my research turned out. However, I would like to change my dataset to a more realistic one where the dataset is small and much simpler than the hospitals' datasets, I was not able to do that because I wasn't able to find any on Kaggle and I didn't have the required resources.
Merits:
Using the mixed method approach enabled me to gather valuable insights. In addition, each step of the research offered valuable insight and there weren't any unnecessary steps. And the research was comprehensive as multiple aspects were covered and each step is influenced by the other.

Limitations:
The research process was time-consuming as there were steps that I didn't have any expertise in so I had to self-study. However, overall there weren't any major limitations of the research process as all the steps were necessary and I have learned from each one.

Pitfalls:
Possible unalignment between the project findings might affect the overall research findings.

7.2 Alternative Research Methodologies

There are many ways to conduct and complete this research. Starting from the literature review where there are many approaches such as Chronological, Methodological, Thematic, and Theoretical, in my research I have chosen the Chronological and Methodological where they were the most suitable to be used in my research. Furthermore, there are multiple methodological choices that could be used for research I have used the Mixed method simple where I conducted a survey and I have tested multiple machine-learning models. In my research, I could use the Mixed method complex where I can add other research strategies to make my research more comprehensive, or Multimethod qualitative where I could focus on gathering qualitative data more to gather more in-depth insights. In addition, I could change the research strategy where I can conduct interviews which will help cover more aspects and ever reveal unexpected information, or Archival study where I can depend on previous research papers. There are many approaches to analyze the primary data where instead of using Google Forms charts, the gathered data could be saved as a CSV file then use Python and coding to analyze the results.

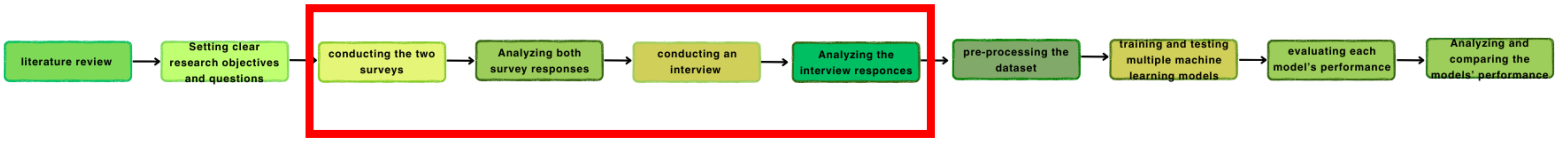
Lessons learned:
When conducting this research I have earned many vital skills. Such as critical thinking, where the analysis process requires significant attention and concentration. As well as earning problem-solving skills, as I have faced many challenges and I was able to overcome them in the end. In addition, I have learned that in order to conduct efficient and valuable research I have to pay attention to the details. One of the major skills that I have earned is social skills where I had to contact experts in data science and doctors, which made me learn how to formally communicate with professionals. As for what I have learned from my research, I have learned how crucial it Is to pay attention to ethical and legal concerns and how data security is a significant aspect that should be highly considered when implementing new systems, especially AI-based systems. In addition, I have learned how Big data could affect our lives as it is present in almost every aspect of it. Finally, I have learned how important regular health checkups are as the earlier the cancer is diagnosed the better the treatment outcomes will be.

7.3 Recommended Actions and Future Considerations

When writing the recommendations, I ensured that they were made based on the analysis resulting in making them effective and relevant, and I have made sure that the recommendations are targeted to the Jordanian hospitals and healthcare sector where the recommendations are actionable and could be followed in Jordan. However, in my recommendations, I have focused more on the ethical and legal concerns and technical challenges if there is more focus on the actual system code such as what are the necessary preprocessing steps, what are the algorithms that could be added to enhance the accuracy like grid search, etc.. This will raise the effectiveness of my recommendations and make them more comprehensive. Thus I recommend that future researchers focus on the code and the system's infostructure.

7.4 Recommended Methodology

I recommend this new approach for the research process where new steps would be added and adjustments will be made to some of the existing steps. The new approach will help raise the effectiveness of the research and make it more comprehensive.



- **Literature Review**
  Adding more research papers that focus more on the opportunities and challenges of using AI and big data for early cancer prediction and discussing the ethical concerns, legal laws, and technical challenges more in-depth. This will help me gather more information and find more gaps to fill which will make the research more advanced and effective.

- **Setting Clear Research Objectives and Questions**
  Expanding the research scope and adding research questions that focus more on details to make the research more comprehensive such as:
  1. How often the system should be maintained and what are the approximate expenses of regular maintenance?
  2. What healthcare staff training is required after implementing the system?

- **Conducting the two Surveys**
  Gathering a decent amount of responses at least 30 responses. to make it more reliable and comprehensive.

  Conducting an additional survey that will be sent to Lung cancer patients in Jordan to gather insights about their concerns regarding being diagnosed with the help of AI and how we can address these concerns. This will help explore solutions to one of the greatest ethical concerns that come along with integrating AI and big data for early lung cancer prediction in Jordan which is obtaining the patients' consent.

- **Analyzing both Survey Responses**
  I would not change anything about my survey analysis process, because my approach of analyzing the survey responses was effective. But in this step, the patients' survey will be analyzed as well.

- **Conducting an Interview**
  This step is added because it will help to gather more detailed insights as interviewing a doctor could reveal overlooked insights and unseen effective information.

- **Analyzing the interview responses**
  This step is added to be able to extract insights from the responses and turn them from only the doctor's answers into valuable insight that we can benefit from.

- **Dataset**

Choosing a dataset that is more realistic and similar to the Jordanian Hospitals' data structure. And ensuring that the data set chosen is big data to have more realistic results and insights.

- **Pre-processing the Dataset**
  Due to changing the dataset this step will be adjusted on depending the requirements of the new dataset.

- **Training and Testing Multiple Machine Learning Models**
  Testing more machine learning models such as SVM and neural network models to find more alternative options and even models that have better performance.

- **Evaluating the models' performance**
  No adjustments will be needed in this step because the used evaluation measures are effective.

- **Analyzing and Comparing the Models' Performance**
  Adding more plots to provide a better explanation of the models' performance.

Resources

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